

SEMINAR III · WEEK 8 · FOUNDATIONS

Equations

What does the unknown represent, and what operation will undo what's happening to it?

Two teaching sessions, not four separate lessons.

SESSION A

One-Step and Two-Step Equations

WHAT IS AN EQUATION?

Think of it as a balance.

Solve using the **inverse operation** of whatever is currently applied to the variable — model one example each of addition, subtraction, multiplication, division.

COMMON MISCONCEPTION

Do I only need to change one side?

No — whatever you do to one side, you must do to the other, or the equation is no longer balanced.

COMMON MISCONCEPTION

$x + 7 = 12$ — multiply both sides?

No — 7 is being added, so subtraction undoes it. Use the inverse of what's actually there.

TWO-STEP EQUATIONS

Undo in reverse order.

$$3x + 5 = 20$$

Undo addition/subtraction first (subtract 5), **then** undo multiplication/division (divide by 3) — not the order the equation was built in.

COMMON MISCONCEPTION

Does $-2x + 9 = 1$ need a different method?

No — same steps. Dividing by a negative just needs the Week 3 sign rules.

EXIT TICKET — SESSION A

1. Solve a two-step equation
2. Solve one with a negative coefficient
3. Identify the correct first step for a given equation

SESSION B

Combine/Distribute First, Then Check & Apply

BEFORE YOU SOLVE

**Combine like terms and
distribute first.**

Combining $2x + 3x$ into $5x$ doesn't change what the equation means. Distributing is technically optional in some cases, but it's more reliable, especially with extra terms outside the parentheses.

COMMON MISCONCEPTION

Does checking a solution mean re-solving it?

*No — substitute the solution back into the **original** equation and confirm both sides are equal.*

COMMON MISCONCEPTION

Once x is found, is the word problem finished?

Not yet — the value has to be interpreted in context, with the right units, before it's a real final answer.

COMMON MISCONCEPTION

If the solution is a fraction or decimal, was there a mistake?

Not necessarily — check the solution for reasonableness in context, don't reject it just for not being a whole number.

ACTIVITY — BUILD, SOLVE, INTERPRET

Turn a scenario into an equation, solve it, then say what it means.

rental fees

wages with a bonus

ticket pricing

EXIT TICKET — SESSION B

Write an equation from a scenario, solve it, and state the answer with correct units and context.

Then: mixed practice, error analysis, Week 8 check, reflection.

TRANSITION TO WEEK 9

Quarter 1 Consolidation

Bringing the quarter's number sense and early algebra skills together before Quarter 2.